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Optimization for Data Science

Frank–Wolfe Variants for the LASSO Problem

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Contents

1 The LASSO Problem and Frank–Wolfe Optimization **2**

1.1 Problem Setup 2

1.2 The Frank–Wolfe Algorithm 2

1.3 Frank–Wolfe Variants 4

1.3.1 Away-Step Frank–Wolfe 4

1.3.2 Pairwise Frank–Wolfe 5

2 Experimental Setup **7**

2.1 Setup and Evaluation 7

2.2 Description of the Datasets 7

3 Analysis on Real-World Data **9**

3.1 Concrete Compressive Strength 9

3.2 California Housing 12

3.3 CT Slice Localization 15

3.4 Dorothea 18

4 Conclusions **21**

A Derivations **22**

A.1 The LASSO Linear Minimization Oracle 22

A.2 Exact Line Search for Classic Frank–Wolfe 22

A.3 Maximum Step Size for the Away-Step Direction 23

A.4 Maximum Step Size for the Pairwise Direction 23

1 The LASSO Problem and Frank–Wolfe Optimization

1.1 Problem Setup

Linear regression models a response variable as a linear function of a set of predictors: given m observations (x_i, y_i), $i = 1, \ldots, m$, with $x_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$, the fitted model is $\hat{y}_i = \sum_{j=1}^{n} \beta_j x_{ij}$ (an explicit intercept β_0 is removed throughout this report for reasons discussed in Chapter 2). Collecting the observations into a design matrix $X \in \mathbb{R}^{m \times n}$ and response vector $y \in \mathbb{R}^m$, ordinary least squares fits β by minimizing $\|X\beta - y\|_2^2$ directly, with no restriction on β. When n is large relative to m, or when many predictors are irrelevant or redundant, this unconstrained fit tends to overfit and can even become ill-posed, since more than one β may fit the training data equally well.

LASSO addresses this by additionally penalizing the size of β specifically, its ℓ_1 norm. The problem solved throughout this report is

$$\min_{\beta \in \mathbb{R}^n} \; f(\beta) = \frac{1}{2m}\|X\beta - y\|_2^2 \qquad \text{subject to} \qquad \|\beta\|_1 \leq t, \tag{1.1}$$

where the radius $t > 0$ sets how much of an ℓ_1 budget the coefficients are allowed. A small t forces most entries of β to become zero, yielding a sparse, easily interpretable model at some cost in fit quality. As t grows, the solution converges to the unconstrained least-squares fit. The feasible set for this problem,

$$\mathcal{C} = \{\beta \in \mathbb{R}^n : \|\beta\|_1 \leq t\}, \tag{1.2}$$

is exactly the ℓ_1-ball of radius t, a polytope whose geometry makes Frank-Wolfe convenient solver for this problem.

1.2 The Frank–Wolfe Algorithm

Frank–Wolfe is a first-order, projection-free method for constrained optimization problems of the form $\min_{x \in \mathcal{C}} f(x)$, where f is convex and differentiable and $\mathcal{C}$ is compact and convex. Rather than taking an unconstrained step and projecting the result back into $\mathcal{C}$, as most gradient-based methods for constrained problems do, FW minimizes a linear approximation of f over the entire feasible set at each iteration and moves toward that minimizer. The procedure is displayed in Algorithm 1.

Algorithm 1 Classic Frank–Wolfe (FW)

Initialize $\beta^{(0)} \in C$
for $k = 0, 1, 2, \ldots$ **do**
  $s_k \leftarrow \arg\min_{s \in C} \nabla f(\beta^{(k)})^\top s$
  $\text{gap}_k \leftarrow -\nabla f(\beta^{(k)})^\top (s_k - \beta^{(k)})$
  if $\text{gap}_k \le \epsilon$ **then stop**
  end if
  $\alpha_k \leftarrow \arg\min_{\alpha \in [0,1]} f(\beta^{(k)} + \alpha(s_k - \beta^{(k)}))$
  $\beta^{(k+1)} \leftarrow \beta^{(k)} + \alpha_k(s_k - \beta^{(k)})$
end for

Applied to the constrained LASSO problem, β takes the role of x, $f(\beta) = \frac{1}{2m}\|X\beta - y\|_2^2$, and the feasible set is $\mathcal{C} = \{\beta \in \mathbb{R}^n : \|\beta\|_1 \le t\}$. The general framework is unchanged. However, specific to LASSO is that every abstract step above admits a closed form, derived in full in Appendix A. The algorithm is initialized at $\beta_0 = 0$, feasible since $\|\beta_0\|_1 = 0 \le t$, with the residual $r_k = X\beta_k - y$ maintained throughout.

The gradient of the objective is

$$\nabla f(\beta_k) = \frac{1}{m} X^\top r_k. \tag{1.3}$$

For a general compact convex set $\mathcal{C}$, the Linear Minimization Oracle (LMO) is the problem $s_k \in \arg\min_{s \in \mathcal{C}} \nabla f(\beta_k)^\top s$. Since this is a linear objective over a convex set, its minimum is attained at a vertex of $\mathcal{C}$, so FW can find the vertex directly instead of computing a projection. For a polytope with a tractable vertex structure, solving the LMO is considerably cheaper than a general convex subproblem. For the LASSO ℓ_1-ball, whose extreme points are $\{\pm t e_i : i = 1, \ldots, n\}$, the LMO reduces to the closed form

$$i_k \in \arg\max_i |\nabla_i f(\beta_k)|, \qquad s_k = -t \, \operatorname{sign}(\nabla_{i_k} f(\beta_k)) \, e_{i_k}, \tag{1.4}$$

derived in Appendix A. This reduces the LMO to a single arg max over n gradient entries, an $O(n)$ operation, and ensures that each s_k has only one nonzero coordinate, so that at most one new coordinate of β_k can become nonzero per iteration.

Given s_k, the FW direction is $d_k = s_k - \beta_k$, and the update $\beta_{k+1} = (1 - \alpha_k)\beta_k + \alpha_k s_k$ remains feasible automatically, as a convex combination of two points already in the ℓ_1-ball. The Frank–Wolfe gap,

$$G(\beta_k) = \max_{s \in \mathcal{C}} -\nabla f(\beta_k)^\top (s - \beta_k) = -\nabla f(\beta_k)^\top d_k, \tag{1.5}$$

satisfies $f(\beta_k) - f^\star \le G(\beta_k)$ by convexity of f, and therefore serves as a stopping criterion: if $G(\beta_k) \le \varepsilon$, β_k is guaranteed to lie within ε of the true optimum.

Because the LASSO objective is quadratic, the step size $\alpha_k = \arg\min_{\alpha \in [0,1]} f(\beta_k + \alpha d_k)$ admits the closed form

$$\alpha_k = \operatorname{clip}\left(-\frac{r_k^\top X d_k}{\|X d_k\|_2^2}, \, 0, \, 1 \right), \tag{1.6}$$

derived in Appendix A. The need for an iterative line search is avoided. The residual is then updated as $r_{k+1} = r_k + \alpha_k X d_k$, reusing the quantity Xd_k already computed for the line search rather than recomputing $X\beta_{k+1} - y$ from scratch.

1.3 Frank–Wolfe Variants

Classic FW is limited to moving toward a newly introduced atom, once several atoms carry weight in β_k, it has no mechanism for reducing the influence of one already in use. Near the boundary of the feasible set this produces a zig-zagging pattern and correspondingly slow convergence. Away-step and pairwise FW address this by introducing a second type of move that acts on the atoms already active, not only introducing new ones.

1.3.1 Away-Step Frank–Wolfe

Both variants require tracking the iterate as an explicit convex combination of atoms,

$$\beta_k = \sum_{v \in \mathcal{S}_k} \lambda_v^{(k)} v, \qquad \lambda_v^{(k)} \geq 0, \qquad \sum_{v \in \mathcal{S}_k} \lambda_v^{(k)} = 1, \tag{1.7}$$

as opposed to only keeping the vector β_k, because a given β_k does not in general determine a unique decomposition into atoms. Each LASSO atom is stored as a pair (i, σ) with $\sigma \in \{-1, +1\}$ representing $v = t\sigma e_i$, with a dictionary mapping each active pair to its weight. The method is initialized at the LMO vertex for $\beta_0 = 0$, so the active set begins with a single atom of weight one.

At each iteration, alongside the FW atom s_k, an *away atom* is selected as the currently active atom least favored by the linearization:

$$v_k \in \arg\max_{v \in \mathcal{S}_k} \nabla f(\beta_k)^\top v. \tag{1.8}$$

Two candidate directions are then available,

$$d_k^{FW} = s_k - \beta_k \qquad \text{and} \qquad d_k^{A} = \beta_k - v_k, \tag{1.9}$$

and whichever produces the larger first-order decrease is selected. The stopping criterion however, always uses the standard gap computed from d_k^{FW}, since only this direction carries the certified bound.

The two directions admit different maximum step sizes. A standard FW step is feasible for any $\gamma \in [0, 1]$, so $\gamma_{\max}^{FW} = 1$. The away step is bounded by

$$\gamma_{\max}^{A} = \frac{\lambda_{v_k}}{1 - \lambda_{v_k}}, \tag{1.10}$$

derived in Appendix A. In both of the cases the step size is obtained from the same closed-form line search as classic FW, clipped to whichever $\gamma_{\max}$ applies. Taking the maximum away step drives λ_{v_k} to zero and removes the atom from the active set, a *drop step*. In the code, an atom

is pruned once its weight falls below a small numerical threshold such as 10^{-12}.

Algorithm 2 Away-Step Frank–Wolfe (AFW)

Initialize $\beta^{(0)} \in C$ via the LMO at $\beta = 0$; set $\mathcal{S}_0 = \{\beta^{(0)}\}$, $\lambda_{\beta^{(0)}} = 1$
for $k = 0, 1, 2, \dots$ **do**
  $s_k \leftarrow \arg\min_{s \in C} \nabla f(\beta^{(k)})^\top s$
  $d_k^{FW} \leftarrow s_k - \beta^{(k)}, \quad g_k^{FW} \leftarrow -\nabla f(\beta^{(k)})^\top d_k^{FW}$
  if $g_k^{FW} \le \epsilon$ **then stop**
  end if
  $v_k \leftarrow \arg\max_{v \in \mathcal{S}_k} \nabla f(\beta^{(k)})^\top v$
  $d_k^{A} \leftarrow \beta^{(k)} - v_k, \quad g_k^{A} \leftarrow -\nabla f(\beta^{(k)})^\top d_k^{A}$
  if $g_k^{FW} \ge g_k^{A}$ **then**
    $d_k \leftarrow d_k^{FW}, \quad \gamma_{\max} \leftarrow 1$
  else
    $d_k \leftarrow d_k^{A}, \quad \gamma_{\max} \leftarrow \lambda_{v_k} / (1 - \lambda_{v_k})$
  end if
  $\gamma_k \leftarrow \mathrm{clip}\big(-\nabla f(\beta^{(k)})^\top d_k / \|X d_k\|^2,\, 0,\, \gamma_{\max}\big)$
  $\beta^{(k+1)} \leftarrow \beta^{(k)} + \gamma_k d_k$
  update active weights λ_v according to whether an FW or away step was taken; prune atoms with $\lambda_v \approx 0$
end for

1.3.2 Pairwise Frank–Wolfe

Pairwise FW uses the same LMO, active-set bookkeeping, and away-atom rule as away-step FW, but combines both directions into a single move,

$$d_k^{PFW} = s_k - v_k, \tag{1.11}$$

transferring weight directly from the away atom to the FW atom rather than choosing between the two. Its maximum step size,

$$\gamma_{\max}^{PFW} = \lambda_{v_k}, \tag{1.12}$$

is derived in Appendix A and is strictly tighter than the away-step bound, since the pairwise update rescales no other atom's weight, when the away-step update rescales the entire active set before subtracting from v_k. Line search and pruning however, follow the same procedure as away-step FW, with $[0, \lambda_{v_k}]$ in place of $[0, \gamma_{\max}^{A}]$.

Algorithm 3 Pairwise Frank–Wolfe (PFW)

Initialize $\beta^{(0)} \in C$ via the LMO at $\beta = 0$; set $\mathcal{S}_0 = \{\beta^{(0)}\}$, $\lambda_{\beta^{(0)}} = 1$
for $k = 0, 1, 2, \ldots$ **do**
  $s_k \leftarrow \arg\min_{s \in C} \nabla f(\beta^{(k)})^\top s$
  $G(\beta^{(k)}) \leftarrow -\nabla f(\beta^{(k)})^\top (s_k - \beta^{(k)})$
  if $G(\beta^{(k)}) \le \epsilon$ **then stop**
  end if
  $v_k \leftarrow \arg\max_{v \in \mathcal{S}_k} \nabla f(\beta^{(k)})^\top v$
  $d_k \leftarrow s_k - v_k, \quad \gamma_{\max} \leftarrow \lambda_{v_k}$
  $\gamma_k \leftarrow \mathrm{clip}\big(-\nabla f(\beta^{(k)})^\top d_k / \|X d_k\|^2,\, 0,\, \gamma_{\max}\big)$
  $\beta^{(k+1)} \leftarrow \beta^{(k)} + \gamma_k d_k$
  $\lambda_{v_k} \leftarrow \lambda_{v_k} - \gamma_k, \quad \lambda_{s_k} \leftarrow \lambda_{s_k} + \gamma_k$; prune atoms with $\lambda_v \approx 0$
end for

Table 1.1 summarizes the three methods. All three share the objective, gradient, LMO, and stopping criterion. The direction, the requirement for an explicit active set, and the maximum step size are different for each. All three achieve linear convergence when f is strongly convex and C is polyhedral, which holds here since $\nabla^2 f(\beta) = \frac{1}{m} X^\top X$ is positive definite for $n \le m$ and the ℓ_1-ball is a polytope. The away-step and pairwise directions allow this faster rate to be realized, by correcting the zig-zagging to which classic FW is prone near the boundary of the feasible set.

	Classic FW	**Away-Step FW**	**Pairwise FW**
FW atom	s_k	s_k	s_k
Away atom	none	v_k	v_k
Direction	$s_k - \beta_k$	$s_k - \beta_k$ or $\beta_k - v_k$	$s_k - v_k$
Active set required	no	yes	yes
Maximum step $\gamma_{\max}$	1	1 or $\dfrac{\lambda_{v_k}}{1 - \lambda_{v_k}}$	λ_{v_k}

Table 1.1: Comparison of the three Frank–Wolfe variants implemented in this report.

2 Experimental Setup

2.1 Setup and Evaluation

Features were standardized to zero mean and unit variance before fitting. The LASSO objective was scaled by $1/m$ for numerical stability across datasets of different sizes. Since the FW implementation has no intercept term, the target y was centered on its mean $\bar{y}$ before fitting, with $\bar{y}$ added back to obtain final predictions, $\hat{y} = X\beta + \bar{y}$.

Data was split 70%/15%/15% into training, validation, and test sets. The regularization radius t was selected via grid search: pairwise FW was run for a range of t values, expressed as a fraction of $t_{\max} = \|\beta_{OLS}\|_1$, the unregularized OLS solution, and the value minimizing validation MSE was selected as t_{best}. Final metrics are reported on the held-out test set, using t_{best}. The duality gap tolerance was set to $\epsilon = 10^{-8}$, with a maximum of 5,000 iterations per run.

For each dataset, results are reported as MSE, R^2, and adjusted R^2 (regression) or accuracy, precision, recall, F1, and AUC (classification), alongside wall-clock time, number of iterations to convergence, and the number of non-zero coefficients (NNZ) as a measure of sparsity.

2.2 Description of the Datasets

Four datasets were used to compare the three Frank–Wolfe variants across a range of problem sizes, ordered here from smallest to largest, with the one classification dataset last.

Concrete Compressive Strength (1,005 × 8). The smallest and simplest of the four, after removing 25 duplicate rows: 8 predictors describing a concrete mixture's composition, cement, water, aggregates, superplasticizer, curing age, and so on, used to predict compressive strength in MPa. Its small size and low dimensionality make it a quick, easy-to-interpret baseline against which the other, larger datasets can be compared.

California Housing (20,640 × 8). A regression dataset predicting median house value per district from predictors such as median income, house age, average rooms, population, and location. Same number of features as Concrete, but a much larger sample size, useful for observing how convergence behavior holds up as n grows while p stays small.

CT Slice Localization (53,500 × 384). The largest of the regression datasets, built from histograms of bone structure and air inclusion extracted from axial CT images, predicting how far along the body axis a given slice sits. Both n and p are considerably larger here than in the previous two datasets, giving a harder, more demanding case for testing convergence speed.

Dorothea (1,150 × 100,000). A binary classification dataset from the NIPS 2003 feature selection challenge, used to evaluate the same LASSO solver when its linear output is passed through a sigmoid and thresholded rather than used directly for regression. Chemical compounds are described by 100,000 binary features, half of which are deliberately mean-

ingless "probe" features added by the challenge organizers, and labeled as active or inactive binders to a target receptor. Combining the labeled training and validation splits (the official test labels were never released) gives 1,150 usable instances, with only 112 positive, a genuinely imbalanced dataset. Unlike the three regression datasets, this one has far more features than samples, and the presence of known-irrelevant probe features provides a direct check on whether the LASSO sparsity constraint recovers the informative half of the feature space.

3 Analysis on Real-World Data

3.1 Concrete Compressive Strength

Table 3.1 summarizes the comparison across the three Frank–Wolfe variants at the selected value of t.

Method	MSE	R^2	Adj. R^2	NNZ	Iters	Time (s)
Classic FW	122.9393	0.5105	0.4829	8	5000	0.2109
Away-step FW	123.1571	0.5096	0.4820	8	310	0.0185
Pairwise FW	123.1571	0.5096	0.4820	8	152	0.0081

Table 3.1: Comparison of the Frank–Wolfe variants on Concrete Compressive Strength.

With only 8 predictors, none of the three methods finds any sparsity to exploit at this t: all three keep every single feature (NNZ = 8), shown in Figure 3.1. This is the expected outcome given the dataset's low dimensionality, LASSO's sparsity advantage only becomes meaningful when there are more features to discard than there are useful ones, and with just 8 predictors describing a physical mixture, essentially all of them carry genuine predictive signal.

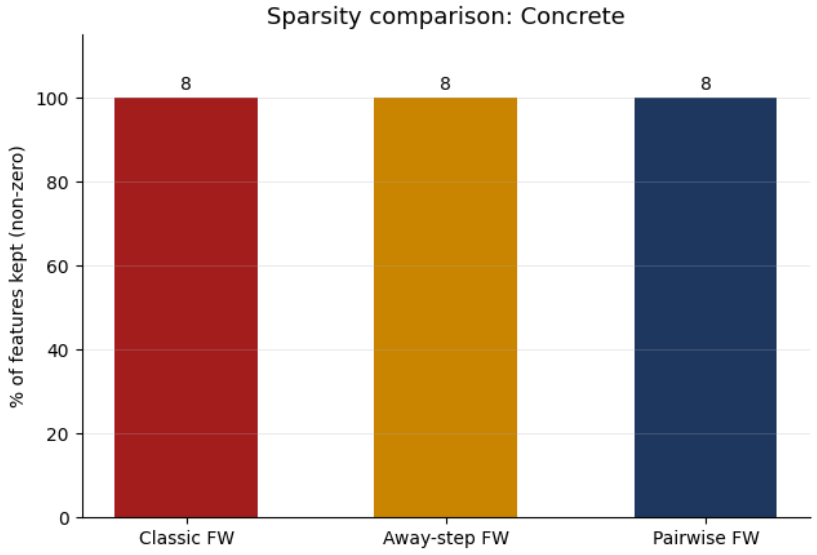


Figure 3.1: Sparsity comparison on Concrete Compressive Strength.

Even without any sparsity difference to separate them, the three variants diverge sharply in convergence speed, shown in Figure 3.2. Classic FW does not reach the duality-gap tolerance within its 5000-iteration budget, ending at a gap of 1.65×10^{-1}, well above 10^{-8}, while away-step FW converges in 310 iterations and pairwise FW in just 152, both reaching gaps below 10^{-8}. This is roughly a 16–33× reduction in iterations, and the wall-clock gap is even larger:

pairwise FW finishes in 0.0081s against classic FW's 0.2109s, close to a 26× speedup, since classic FW's extra iterations more than offset its lower per-iteration cost.

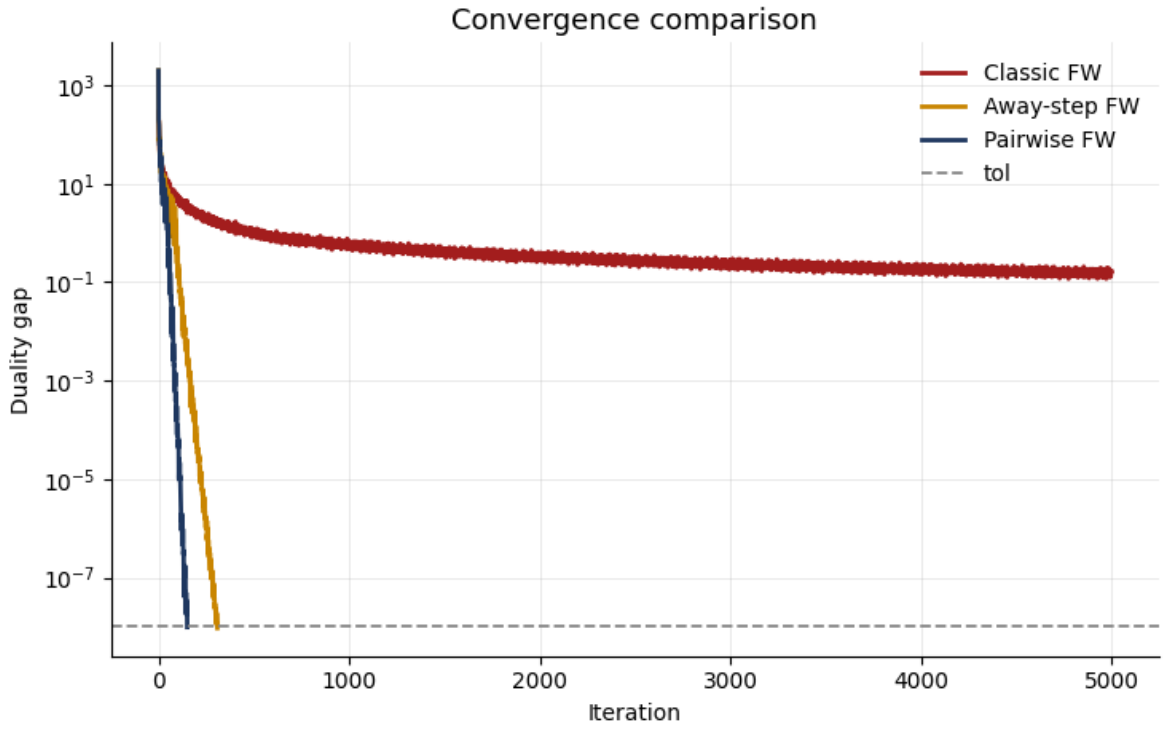


Figure 3.2: Duality gap comparison across the three Frank–Wolfe variants on Concrete Compressive Strength.

Individual convergence curves for each method are shown in Figures 3.3–3.5. Despite the large difference in convergence speed, all three methods reach essentially the same objective value (48.52 for classic FW, 48.45 for both away-step and pairwise), confirming that classic FW's failure to reach the certified tolerance within budget is a matter of convergence speed rather than solution quality: its objective and predictive performance have already stabilized to nearly the same values as the two variants that did converge, well before its duality gap has caught up.

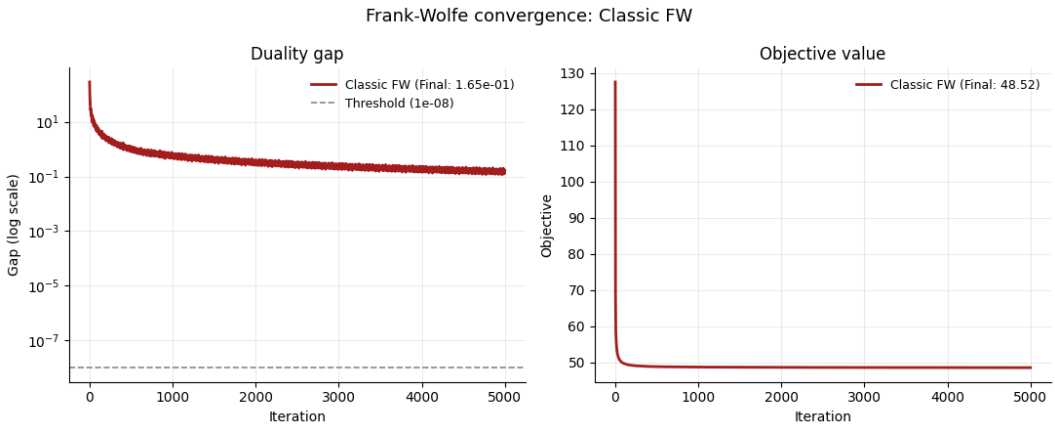


Figure 3.3: Duality gap and objective value for Classic FW on Concrete Compressive Strength.

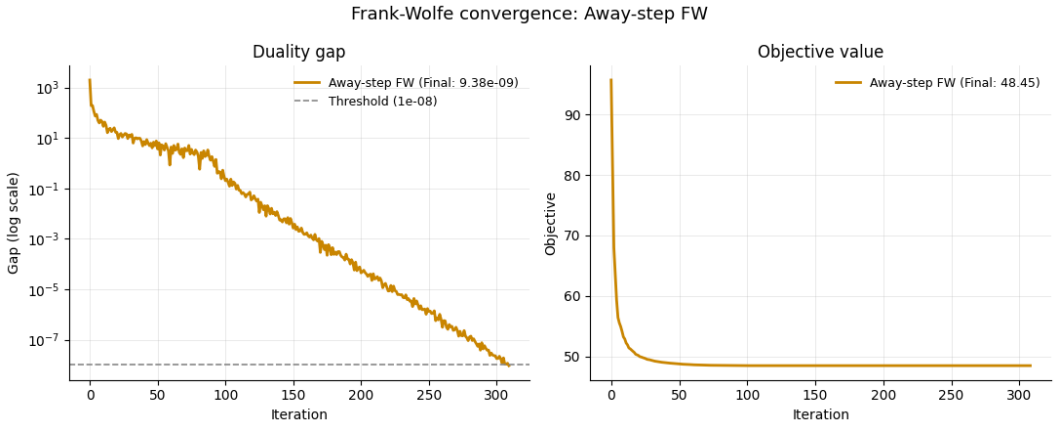


Figure 3.4: Duality gap and objective value for Away-step FW on Concrete Compressive Strength.

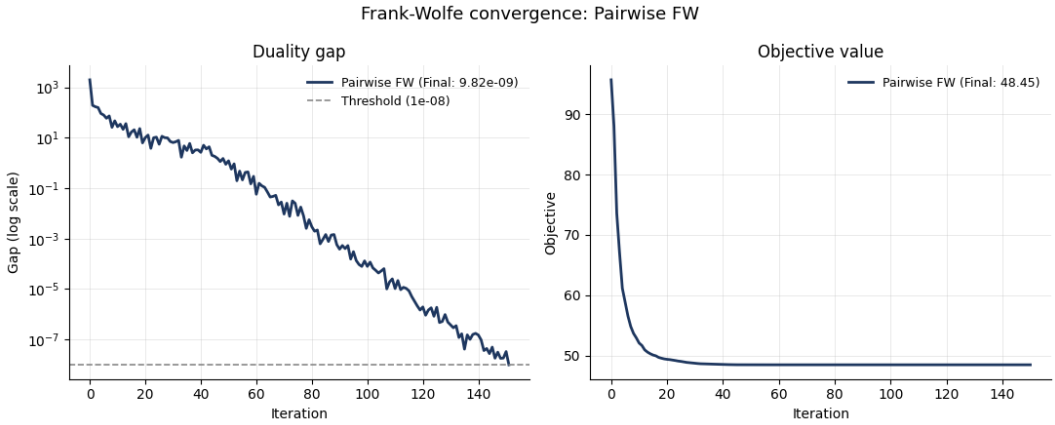


Figure 3.5: Duality gap and objective value for Pairwise FW on Concrete Compressive Strength.

3.2 California Housing

Table 3.2 summarizes the comparison across the three Frank–Wolfe variants.

Method	MSE	R^2	Adj. R^2	NNZ	Iters	Time (s)
Classic FW	0.5201	0.6066	0.6056	8	5000	1.3141
Away-step FW	0.5203	0.6066	0.6055	8	1267	0.3758
Pairwise FW	0.5203	0.6066	0.6055	8	829	0.2305

Table 3.2: Comparison of the Frank–Wolfe variants on California Housing.

As with Concrete Compressive Strength, all 8 available features remain nonzero across all three methods at this value of t (Figure 3.6), so sparsity does not differentiate the variants here either. With so few predictors relative to the sample size, there is little redundant or irrelevant structure for LASSO to discard.

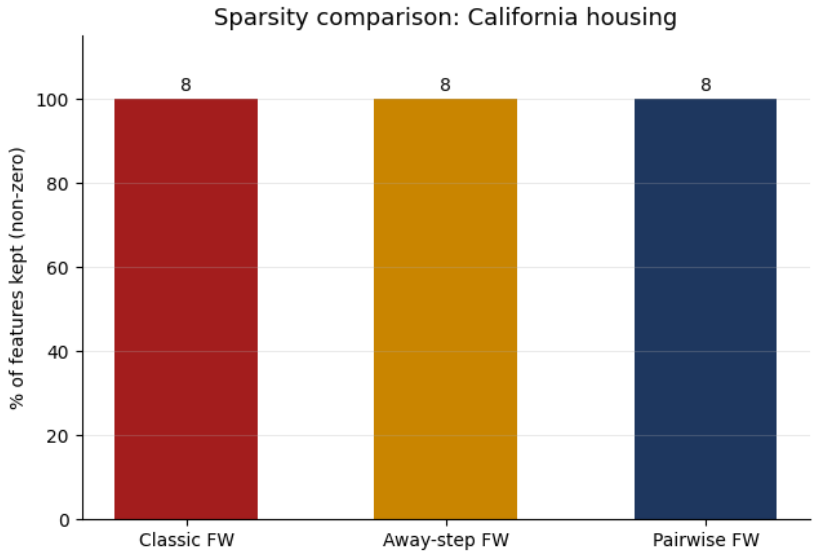


Figure 3.6: Sparsity comparison on California Housing.

Convergence speed again separates the three variants sharply, and more clearly than on Concrete. Classic FW does not reach the duality-gap tolerance within its 5000-iteration budget, ending at 7.60×10^{-5}, roughly four orders of magnitude above 10^{-8}. Away-step FW converges in 1267 iterations and pairwise FW in 829, both certifying convergence well within budget, a 4–6× reduction in iterations relative to classic FW's (incomplete) run. The wall-clock advantage is comparable: pairwise FW finishes in 0.2305s against classic FW's 1.3141s, roughly a 5.7× speedup.

Individual convergence curves are shown in Figures 3.8–3.10. As with Concrete, the objective value has already stabilized to essentially the same point for all three methods (0.26 in each case) long before classic FW's duality gap catches up, reinforcing that its failure to reach the certified tolerance reflects slow convergence near the optimum rather than a materially worse solution.

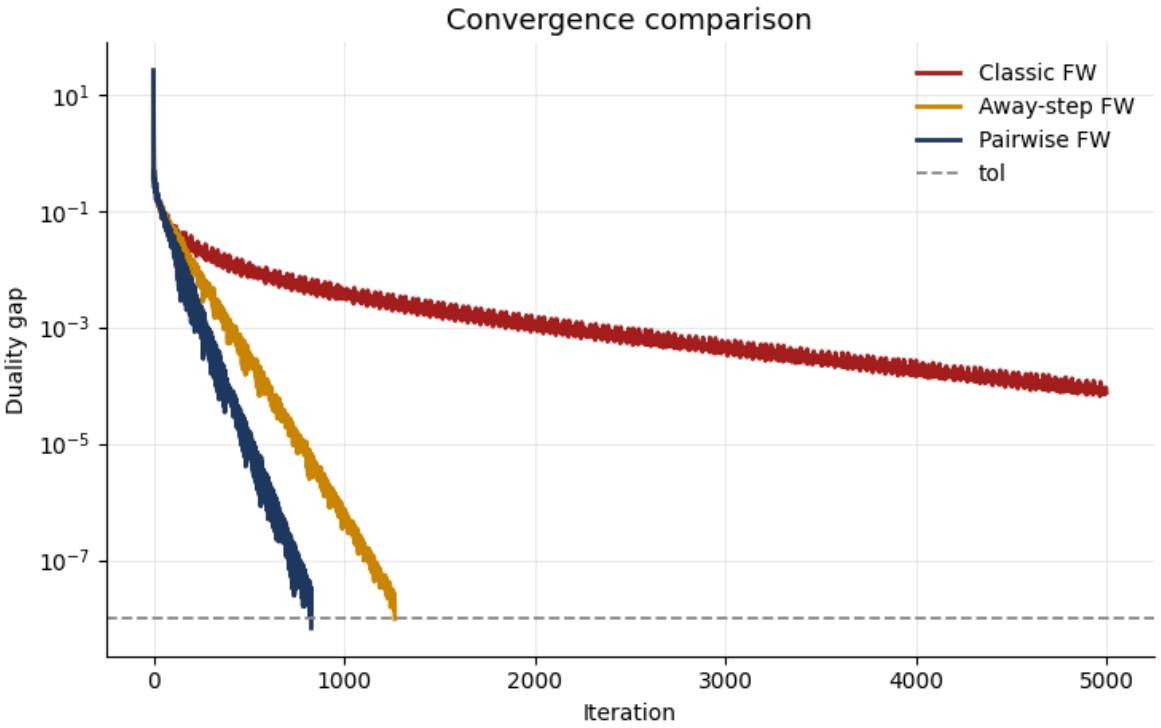


Figure 3.7: Duality gap comparison across the three Frank–Wolfe variants on California Housing.

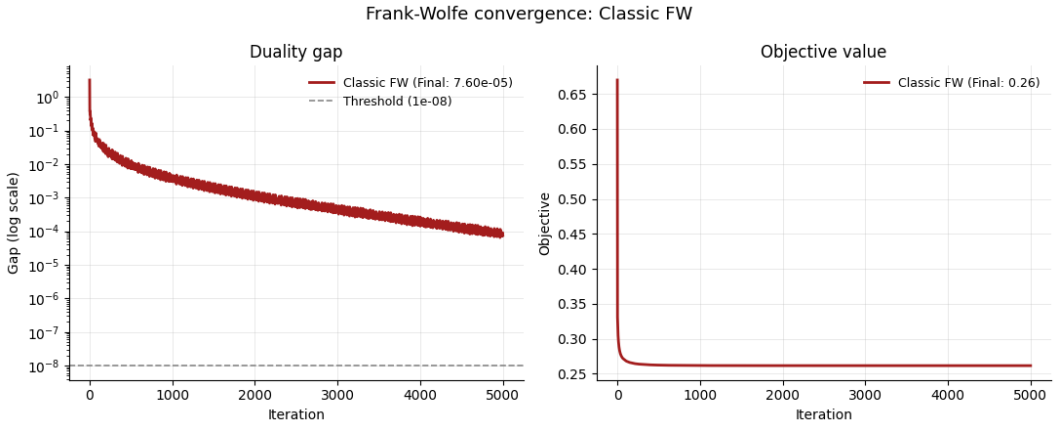


Figure 3.8: Duality gap and objective value for Classic FW on California Housing.

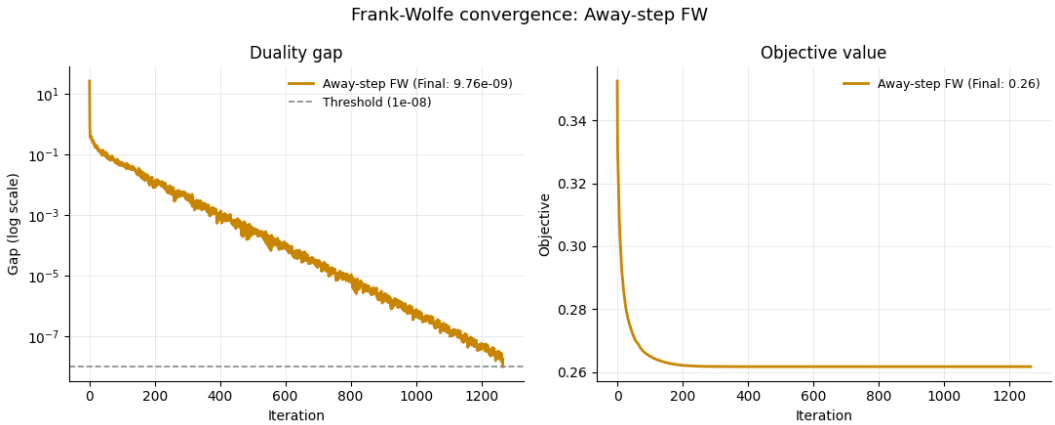


Figure 3.9: Duality gap and objective value for Away-step FW on California Housing.

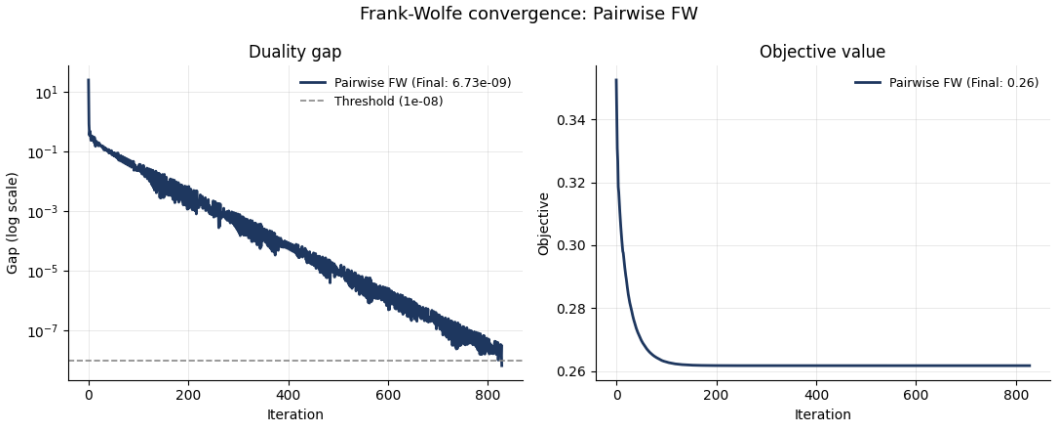


Figure 3.10: Duality gap and objective value for Pairwise FW on California Housing.

3.3 CT Slice Localization

Table 3.3 summarizes the comparison across the three Frank–Wolfe variants.

Method	MSE	R^2	Adj. R^2	NNZ	Iters	Time (s)
Classic FW	69.1634	0.8625	0.8571	303	5000	111.4859
Away-step FW	68.8388	0.8632	0.8571	340	5000	114.9801
Pairwise FW	68.8327	0.8632	0.8570	347	5000	113.4284

Table 3.3: Comparison of the Frank–Wolfe variants on CT Slice Localization.

None of the three variants reaches the duality-gap tolerance within the 5000-iteration budget on this dataset, unlike the two smaller regression datasets. This is consistent with the sparsity pattern in Figure 3.11: the selected value of t admits a solution using 303 to 347 of the 384 available features, close to the full feature set, leaving comparatively little of the sparse structure that away-step and pairwise FW are designed to exploit. Accordingly, all three methods converge at broadly similar, slow rates, shown in Figure 3.12, in contrast to the clear separation observed on Concrete Compressive Strength and California Housing.

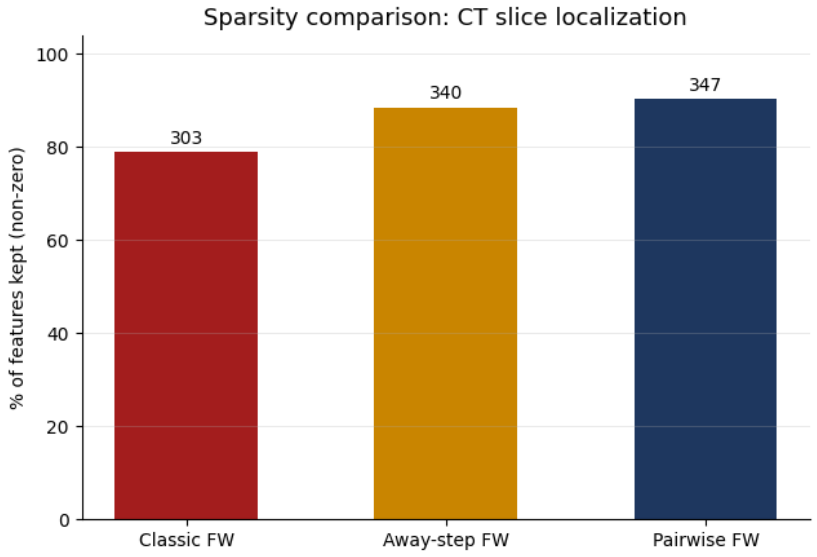


Figure 3.11: Sparsity comparison on CT Slice Localization.

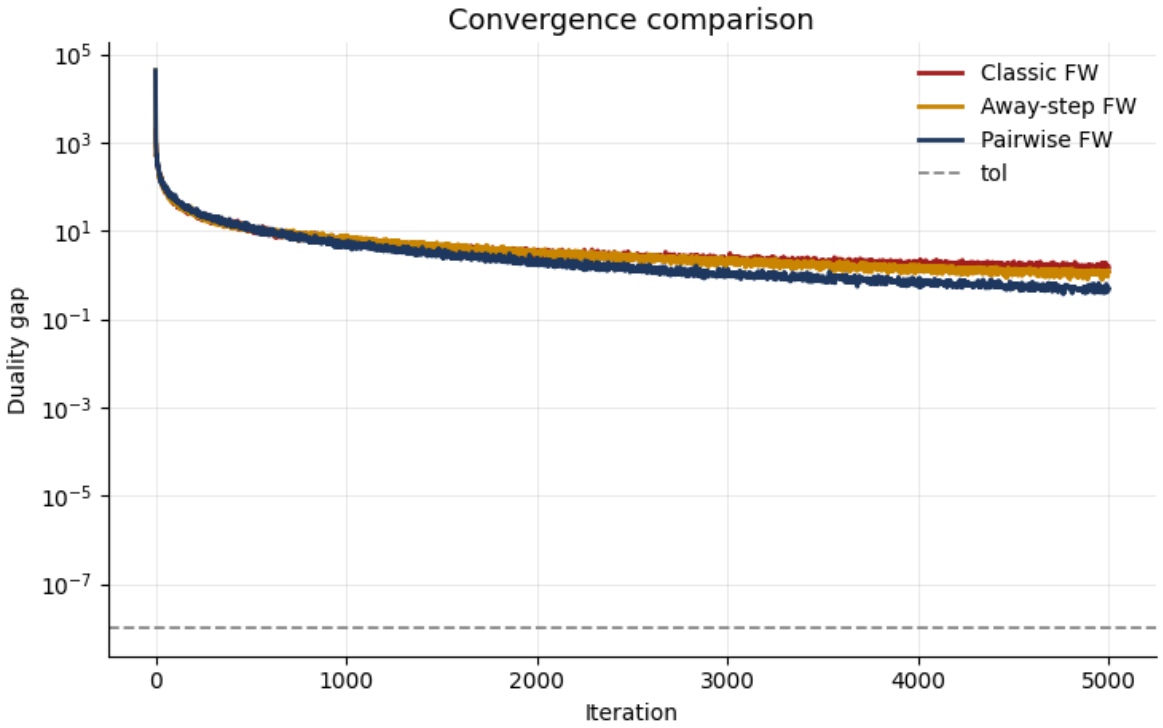


Figure 3.12: Duality gap comparison across the three Frank–Wolfe variants on CT Slice Localization.

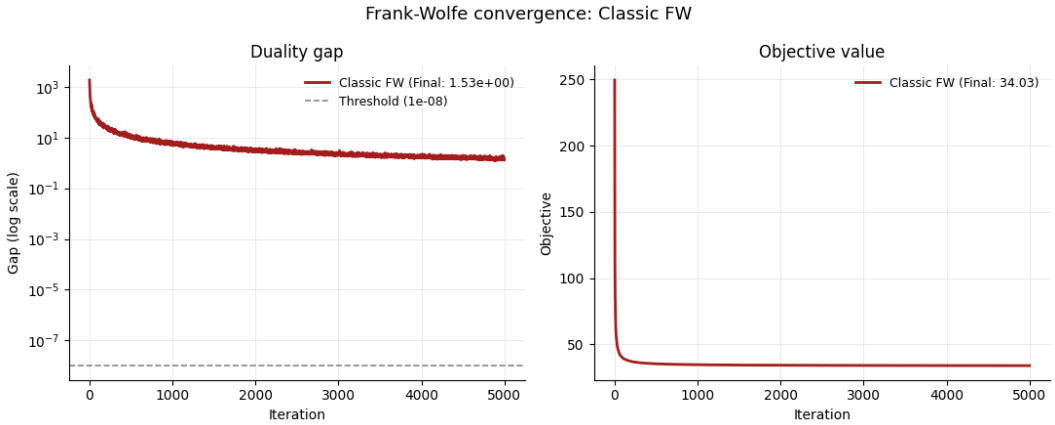


Figure 3.13: Duality gap and objective value for Classic FW on CT Slice Localization.

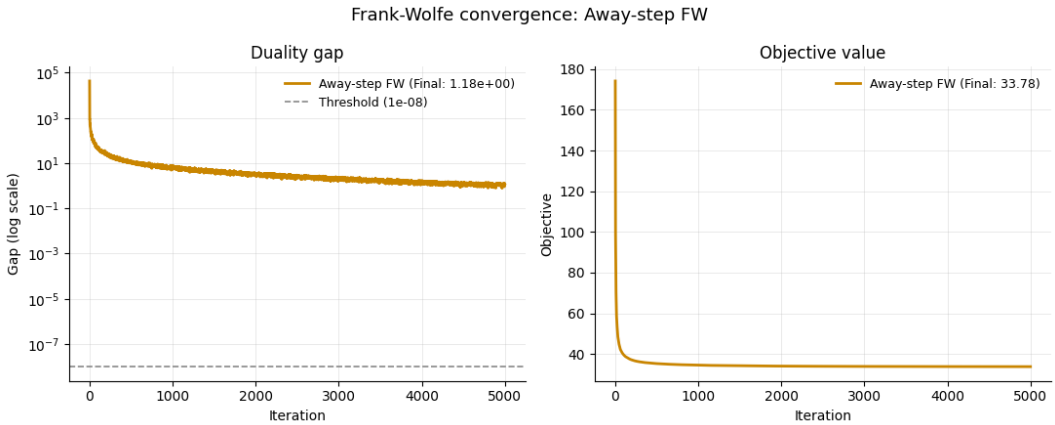


Figure 3.14: Duality gap and objective value for Away-step FW on CT Slice Localization.

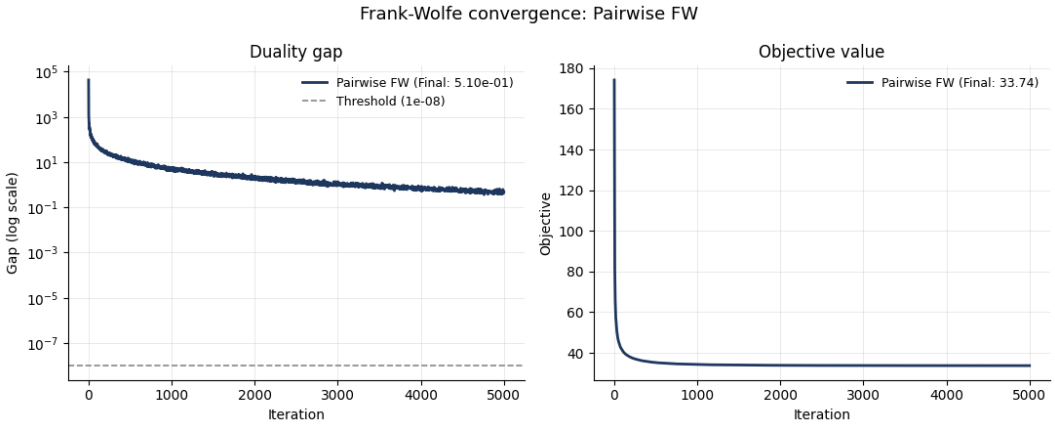


Figure 3.15: Duality gap and objective value for Pairwise FW on CT Slice Localization.

3.4 Dorothea

Table 3.4 summarizes classification performance and convergence across the three variants.

Method	Acc	Prec	Rec	F1	AUC	NNZ	Iters	Time (s)
Classic FW	0.9133	0.5455	0.7059	0.6154	0.8484	63	5000	623.8012
Away-step FW	0.9133	0.5455	0.7059	0.6154	0.8477	58	2684	313.6973
Pairwise FW	0.9133	0.5455	0.7059	0.6154	0.8484	58	1077	121.3025

Table 3.4: Comparison of the Frank–Wolfe variants on Dorothea.

All three methods reach essentially identical classification performance (accuracy, precision, recall, and F1 are equal to four decimal places, with AUC differing only in the fourth digit), again confirming they are solving the same underlying problem. Given the strong class imbalance in this dataset, F1 (0.6154) and AUC (≈ 0.848) are the more informative figures than accuracy, and both indicate a genuinely useful classifier rather than one simply favoring the majority class.

The sparsity result here is the most striking of the four datasets. Out of 100,000 available features, half of which were deliberately uninformative "probe" features added by the challenge organizers, all three variants select only 58 to 63 nonzero coefficients (Figure 3.16), well under 0.1% of the feature space. With a feature space engineered to be half noise, the fitted model relies on a tiny fraction of the available predictors, indicating that the constraint is filtering out irrelevant structure rather than merely shrinking coefficients uniformly.

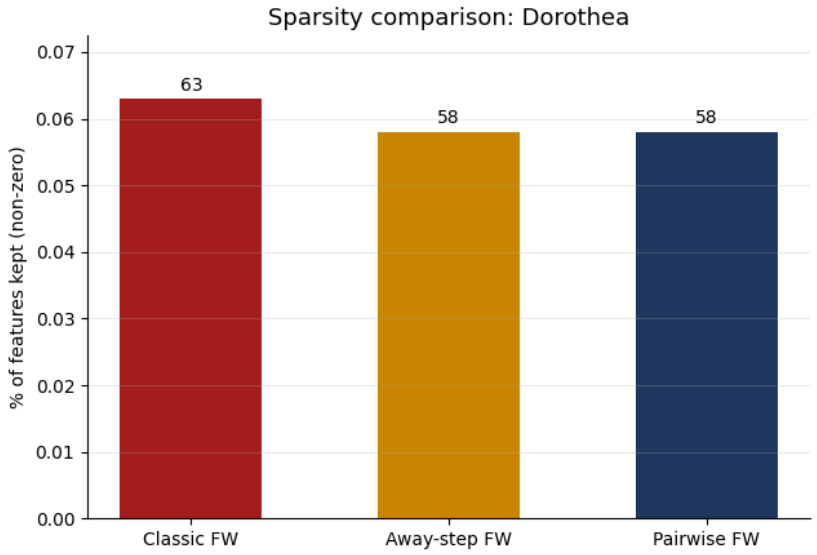


Figure 3.16: Sparsity comparison on Dorothea. The y-axis is in percent of the full 100,000-feature space; all three methods select under 0.1% of available features.

Convergence speed again favors the two variants over classic FW, though the gap here is measured in real time rather than iteration count alone, given the cost of each iteration at this dimensionality. Classic FW does not reach the duality-gap tolerance within its 5000-

iteration budget, ending at 4.93×10^{-5}, while away-step FW converges in 2684 iterations and pairwise FW in 1077, both certifying convergence below 10^{-8}. Pairwise FW is roughly 2.5× faster than away-step FW in iterations and correspondingly faster in wall-clock time (121.3s versus 313.7s), and more than 5× faster than classic FW's incomplete 623.8s run. At 100,000 features, each iteration is substantially more expensive than on the regression datasets, so the absolute time savings from faster convergence are considerably larger here even though the relative iteration-count speedups are broadly consistent with the other three datasets.

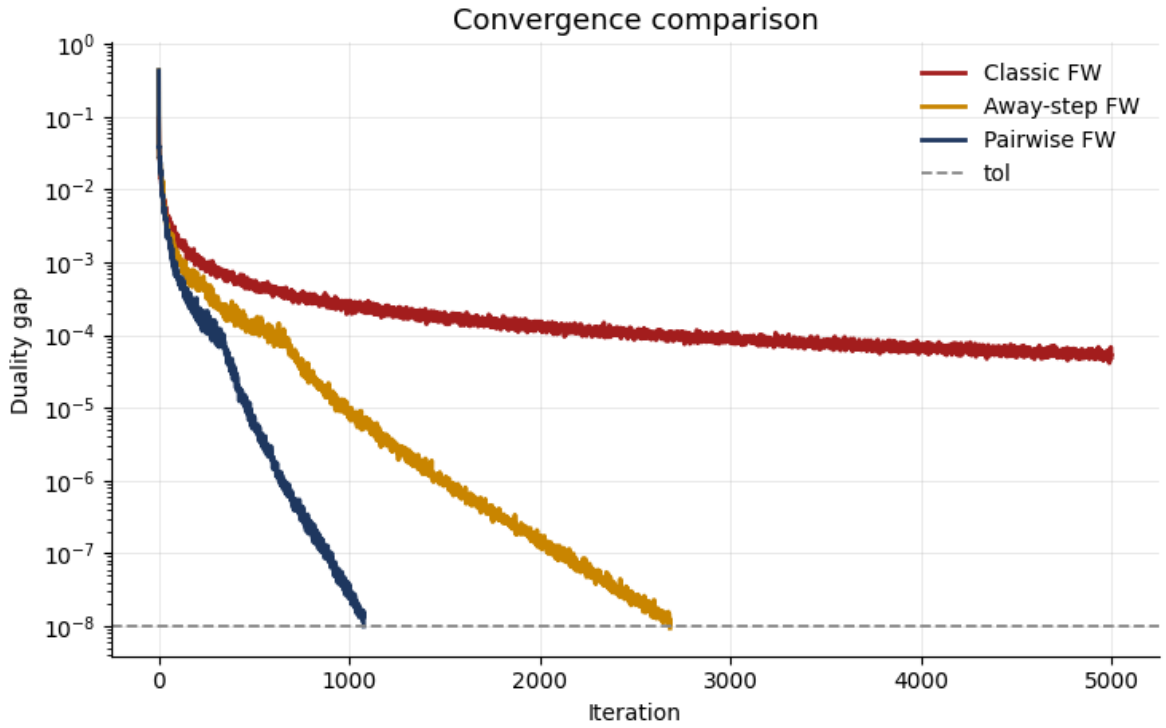


Figure 3.17: Duality gap comparison across the three Frank–Wolfe variants on Dorothea.

As with the regression datasets, the objective value stabilizes to nearly the same point for all three methods (0.02 in each case) well before classic FW's duality gap has caught up, shown in Figures 3.18–3.20, reinforcing that its non-convergence within budget is a matter of convergence speed rather than a meaningfully different or worse solution.

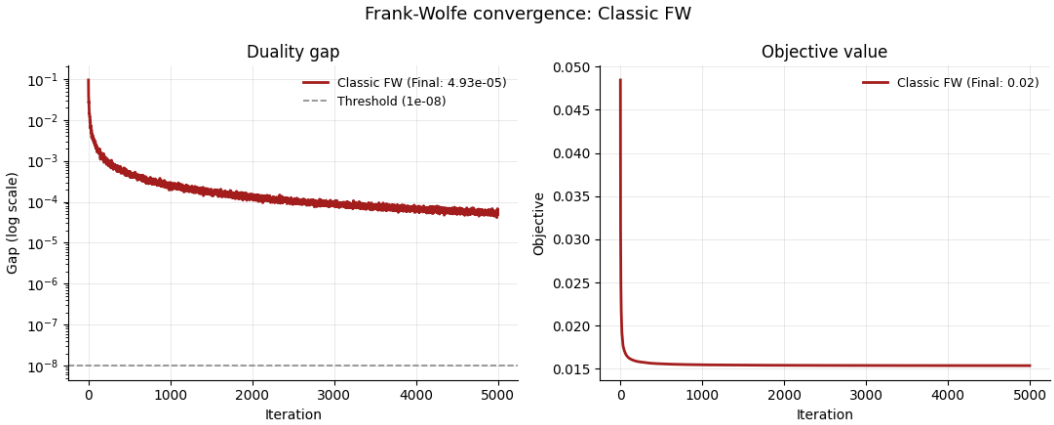


Figure 3.18: Duality gap and objective value for Classic FW on Dorothea.

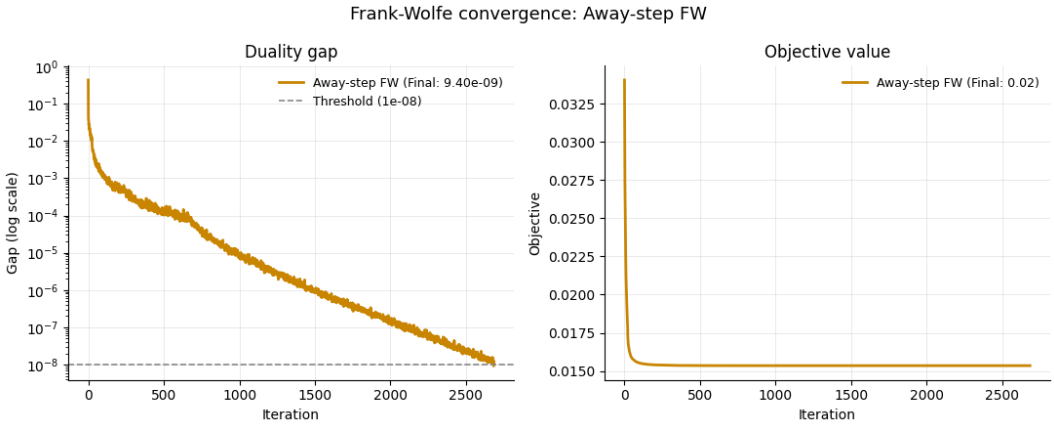


Figure 3.19: Duality gap and objective value for Away-step FW on Dorothea.

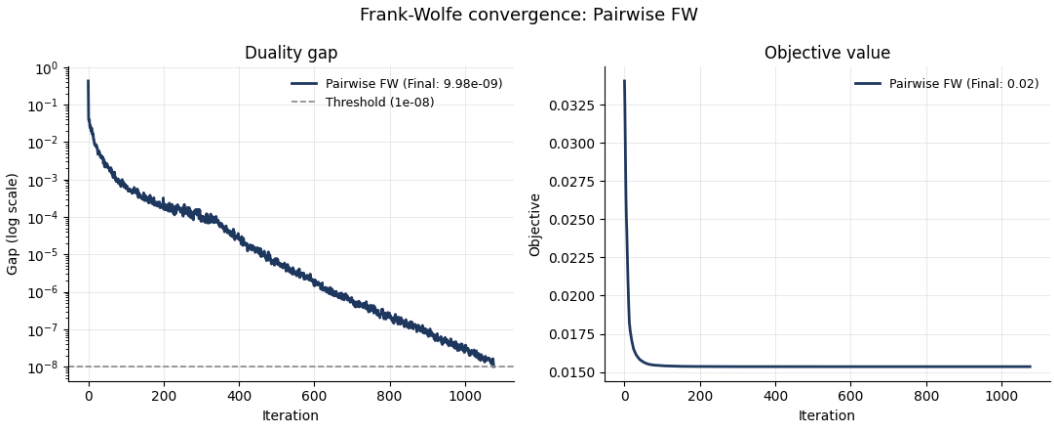


Figure 3.20: Duality gap and objective value for Pairwise FW on Dorothea.

4 Conclusions

Across all four datasets, classic Frank–Wolfe never reached the certified duality-gap tolerance within its 5000-iteration budget, its slow, zig-zagging convergence near the constrained optimum showing up consistently regardless of problem size or task type. Away-step and pairwise FW both correct this by allowing movement away from poorly-weighted active atoms rather than only ever moving toward a new vertex, and both certified convergence well within budget on every dataset tested. Pairwise FW was consistently the fastest of the three, needing roughly 2–33× fewer iterations than classic FW depending on the dataset, and a correspondingly large wall-clock advantage that grew even larger on the highest-dimensional dataset, Dorothea, where the cost of each iteration itself is substantial. Away-step FW sits between the two, correcting much of classic FW's inefficiency without matching pairwise FW's speed.

Despite these large differences in convergence speed, all three variants converge to solutions of essentially identical quality on every dataset, the same MSE, R^2, and sparsity pattern for the regression tasks, and the same accuracy, F1, and AUC for the classification task, confirming they solve the same convex optimization problem via different iterate paths. The one place a difference did show up was Dorothea, where the deliberately engineered feature space, half genuine predictors and half meaningless probes, resulted in fewer than 65 of 100,000 available features being kept by any of the three variants, indicating that the constraint recovers informative structure rather than simply shrinking every coefficient uniformly.

A Derivations

This appendix derives, in full, the closed forms stated without proof in Chapter 1: the LASSO Linear Minimization Oracle, the exact line search, and the maximum step sizes for the away-step and pairwise directions.

A.1 The LASSO Linear Minimization Oracle

The general LMO is $s_k \in \arg\min_{s\in\mathcal{C}} \nabla f(\beta_k)^\top s$. For LASSO, $\mathcal{C}$ is the ℓ_1-ball of radius t, so

$$s_k \in \arg\min_{\|s\|_1 \le t} \nabla f(\beta_k)^\top s. \tag{A.1}$$

Writing $g = \nabla f(\beta_k)$, for any feasible s,

$$g^\top s = \sum_{i=1}^n g_i s_i \;\ge\; -\sum_{i=1}^n |g_i||s_i| \;\ge\; -\|g\|_\infty \sum_{i=1}^n |s_i| = -\|g\|_\infty \|s\|_1 \;\ge\; -t\|g\|_\infty, \tag{A.2}$$

so $-t\|g\|_\infty$ is a lower bound on the LMO value. Setting $i_k \in \arg\max_i |g_i|$, so that $|g_{i_k}| = \|g\|_\infty$, and

$$s_k = -t\ \mathrm{sign}(g_{i_k})\, e_{i_k}, \tag{A.3}$$

gives a feasible point ($\|s_k\|_1 = t$) with

$$g^\top s_k = g_{i_k}\,[-t\ \mathrm{sign}(g_{i_k})] = -t|g_{i_k}| = -t\|g\|_\infty, \tag{A.4}$$

which attains the bound exactly, so s_k solves the LMO. In terms of the gradient, this gives the closed form used throughout Chapter 1,

$$i_k \in \arg\max_i |\nabla_i f(\beta_k)|, \qquad s_k = -t\ \mathrm{sign}(\nabla_{i_k} f(\beta_k))\, e_{i_k}. \tag{A.5}$$

A.2 Exact Line Search for Classic Frank–Wolfe

The line search solves $\alpha_k = \arg\min_{\alpha\in[0,1]} f(\beta_k + \alpha d_k)$. Substituting $\beta_k + \alpha d_k$ into f and writing $r_k = X\beta_k - y$,

$$\begin{aligned} f(\beta_k + \alpha d_k) &= \frac{1}{2m}\|r_k + \alpha X d_k\|_2^2 \\ &= \frac{1}{2m}\left[r_k^\top r_k + 2\alpha\, r_k^\top X d_k + \alpha^2 (Xd_k)^\top (Xd_k)\right], \end{aligned} \tag{A.6}$$

a quadratic in α. Setting the derivative with respect to α to zero,

$$\frac{1}{m}\left[r_k^\top X d_k + \alpha (X d_k)^\top (X d_k)\right] = 0, \tag{A.7}$$

yields

$$\alpha_k^\star = -\frac{r_k^\top X d_k}{\|X d_k\|_2^2} \tag{A.8}$$

(the factor $1/m$ cancels, as it multiplies every term). This is the unconstrained minimizer along the line through β_k in direction d_k; imposing $\alpha_k \in [0, 1]$ gives the constrained step size

$$\alpha_k = \text{clip}\left(-\frac{r_k^\top X d_k}{\|X d_k\|_2^2},\, 0,\, 1\right). \tag{A.9}$$

A.3 Maximum Step Size for the Away-Step Direction

The away-step update is $\beta_{k+1} = \beta_k + \gamma(\beta_k - v_k) = (1+\gamma)\beta_k - \gamma v_k$. Substituting the convex decomposition $\beta_k = \sum_{u \in \mathcal{S}_k} \lambda_u u$,

$$\beta_{k+1} = (1+\gamma) \sum_{u \in \mathcal{S}_k} \lambda_u u - \gamma v_k, \tag{A.10}$$

so every atom other than v_k has its weight rescaled to $(1+\gamma)\lambda_u$, while v_k's own weight becomes

$$\lambda_{v_k}^{\text{new}} = (1+\gamma)\lambda_{v_k} - \gamma. \tag{A.11}$$

Feasibility requires this to remain nonnegative,

$$(1+\gamma)\lambda_{v_k} - \gamma \geq 0 \quad \Longleftrightarrow \quad \lambda_{v_k} - \gamma(1 - \lambda_{v_k}) \geq 0 \quad \Longleftrightarrow \quad \gamma(1 - \lambda_{v_k}) \leq \lambda_{v_k}, \tag{A.12}$$

giving, for $\lambda_{v_k} < 1$,

$$\gamma_{\max}^A = \frac{\lambda_{v_k}}{1 - \lambda_{v_k}}. \tag{A.13}$$

A.4 Maximum Step Size for the Pairwise Direction

The pairwise update, $\beta_{k+1} = \beta_k + \gamma(s_k - v_k)$, changes only the weight of v_k,

$$\lambda_{v_k}^{\text{new}} = \lambda_{v_k} - \gamma, \tag{A.14}$$

so feasibility ($\lambda_{v_k} - \gamma \geq 0$) gives directly

$$\gamma_{\max}^{PFW} = \lambda_{v_k}. \tag{A.15}$$

This bound is tighter than the away-step bound because the pairwise update involves no rescaling of the remaining active weights; the away-step bound's $(1 - \lambda_{v_k})$ term arises specifically from the $(1+\gamma)$ rescaling of the full decomposition, which the pairwise update does not

perform.